

SOLIDWORKS API (.NET VBA C#) — Reference (128/337): IGtol (part 1 of 3)

SOLIDWORKS API Documentation (converted from CHM)

IGtol (part 1 of 3) (Type)

SOLIDWORKS API Help

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IGtol Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGtol Interface

Allows you to get and set geometric tolerance (GTol) parameters.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IGtol
```

Visual Basic (Usage)



```
Dim instance As IGtol
```

C#

```
public interface IGtol
```

C++/CLI

```
public interface class IGtol
```

Visual Basic for Applications (VBA) Syntax

See [Gtol](#).

Example

[Get Text Items in GTol Frame \(VBA\)](#)
[Get Text Items in GTol Frame \(VB.NET\)](#)
[Get Text Items in GTol Frame \(C#\)](#)
[Get GTol Witness Line \(VBA\)](#)
[Get GTol Witness Line \(VB.NET\)](#)
[Get GTol Witness Line \(C#\)](#)
[Set Text in Datum Tags and GTols \(VBA\)](#)
[Set Text in Datum Tags and GTols \(VB.NET\)](#)
[Set Text in Datum Tags and GTols \(C#\)](#)
[Insert GTol \(C#\)](#)
[Insert GTol \(VB.NET\)](#)
[Insert GTol \(VBA\)](#)

Accessors

[IAnnotation::GetSpecificAnnotation](#) and [IAnnotation::IGetSpecificAnnotation](#)

[IDrawingDoc::EditSelectedGtol](#) and [IDrawingDoc::IEditSelectedGtol](#)

[IDrawingDoc::NewGtol](#) and [IDrawingDoc::INewGtol](#)

[IGtol::GetNextGTOL](#) and [IGtol::IGetNextGTOL](#)

[IModelDoc2::InsertGtol](#) and [IModelDoc2::IInsertGtol](#)

[IView::GetFirstGTOL](#) and [IView::IGetFirstGTOL](#)

[IView::GetGTols](#) and [IView::IGetGTols](#)

Access Diagram

[Gtol](#)

See Also

[IGtol Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IGtol Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGtol Interface

For a list of all members of this type, see [IGtol members](#).

Public Methods

Name	Description
AddFrame	Adds a frame at the end of the list of this Gtol's frames.
CanConvertFormat	Gets whether this Gtol can be converted to the current format.
ConvertFormat	Converts this Gtol to the current format.
DeleteBelowFrameTextAt	Deletes the specified text line in the below frame of this Gtol.
DeleteFrame	Deletes the specified frame from this Gtol.
GetAllAround	Gets whether this Gtol uses an All Around leader.
GetAllAroundThisSide	Gets whether this Gtol uses an All Around This Side leader.
GetAllOver	Gets whether this Gtol uses an All Over leader.
GetAllOverThisSide	Gets whether this Gtol uses an All Over This Side leader.
GetAnnotation	Gets the annotation for this specific Gtol.
GetArcAtIndex	Gets information about the specified arc.
GetArcCount	Gets the number of arcs in this Gtol.
GetArrowHeadAtIndex	Gets information on the specified arrow head in this Gtol.
GetArrowHeadCount	Gets the number of arrow heads in the Gtol.
GetArrowHeadInfo	Gets an array of doubles that describe the geometry of the arrowhead on the far end of the leader line.
GetAttachPos	Gets the attachment point of the Gtol.
GetBelowFrameTextAt	Gets the specified line of text in this Gtol.
GetBelowFrameTextLineCount	Gets the number of text lines in the below frame of this Gtol.
GetBetweenTwoPoints	Gets whether the between two points symbol is being used.
GetBetweenTwoPointsText	Gets the text used in the between two points symbol.
GetCompositeFrame	Obsolete. Superseded by IGtol.GetCompositeFrame2 .
GetCompositeFrame2	Gets whether the Gtol frame with the specified index is part of a composite frame.
GetDatumIdentifier	Gets the datum identifier.
GetDisplayDualDimensionInRangeValues	Gets whether to display dual units in dimension range values.
GetFontInfo	Gets with the font information for this Gtol.
GetFormat	Gets the format type of this Gtol.
GetFrame	Gets the Gtol frame at the specified index.
GetFrameCount	Gets the number of frames in this Gtol symbol.
GetFrameDiameterSymbols	Gets whether diameter symbols are displayed for the specified frame.
GetFrameSymbols	Obsolete. Superseded by IGtol.GetFrameSymbols2 .
GetFrameSymbols2	Obsolete. Superseded by IGtol.GetFrameSymbols3 .
GetFrameSymbols3	Gets the symbols for the specified frame.
GetFrameValues	Gets an array that describes the text that appears in each of the fields of the specified Gtol frame.
GetFromTo	Gets whether the From-To symbol is present in this Gtol.
GetFromToText	Gets the From-To text of this Gtol.

		
	GetHeight	Gets the GTol height.
	GetLeaderAtIndex	Obsolete. Superseded by IGtol::GetLeaderAtIndex2 .
	GetLeaderAtIndex2	Gets information about the specified leader on this GTol.
	GetLeaderCount	Gets the number of leaders on this GTol.
	GetLeaderInfo	Gets information describing the layout of the GTol leader lines.
	GetLeaderSide	Gets where the leader attaches to the symbol.
	GetLineAtIndex	Gets the start and end information for the specified line.
	GetLineCount	Gets the number of linea in this GTol.
	GetNextGTOL	Gets the next GTol.
	GetPTZHeight	Obsolete. Superseded by IGtol::GetPTZHeight2 .
	GetPTZHeight2	Gets the projected tolerance zone for the specified frame and tolerance in this GTol.
	GetSymArcs	Gets an array that defines all arcs in the symbol.
	GetSymCircles	Gets an array that defines all circles in the symbol.
	GetSymDesc	Gets the specified symbol description.
	GetSymEdgeCounts	Gets an array of the number of lines, arcs and circles in the specified symbol.
	GetSymLines	Gets an array that defines all lines in the symbol.
	GetSymName	Gets the symbol name based on the specified index number that SOLIDWORKS uses to identify the symbol.
	GetSymText	Gets the symbol text.
	GetSymTextPoints	Gets the text points for the specified symbol.
	GetText	Gets the specified text part of this GTol.
	GetTextAngleAtIndex	Gets the text angle for the specified piece of text in this GTol.
	GetTextAtIndex	Gets the specified text from this GTol.
	GetTextCount	Gets the number of text items in the GTol.
	GetTextFont	Gets the font for this GTol.
	GetTextFormat	Obsolete. Superseded by IAnnotation::GetTextFormat .
	GetTextHeightAtIndex	Gets the text height for the specified piece of text in this GTol.
	GetTextInvertAtIndex	Gets the specified text item's invert flag.
	GetTextPoint	Gets the GTol text reference point, which is the upper-left corner of the bounding rectangle.
	GetTextPositionAtIndex	Gets the text item's offset relative to note's text point.
	GetTextRefPositionAtIndex	Gets the reference position for the specified text item in this GTol (for example, upper-left, lower-left, center).
	GetTriangleAtIndex	Gets the triangle at the specified index.
	GetTriangleCount	Gets the number of triangles in this GTol.
	GetUseDocTextFormat	Obsolete. Superseded by IAnnotation::GetUseDocTextFormat .
	GetWitnessLine	Gets the extension line that connects the leader of this geometric tolerance with the entity being dimensioned.
	HasExtraLeader	Gets whether the GTol has a bent leaderline or a straight leaderline.
	IGetAnnotation	Gets the annotation for this specific GTol.
	IGetArcAtIndex	Gets information for the specified arc.
	IGetArrowHeadAtIndex	Gets information on the specified arrow head in this GTol.
	IGetArrowHeadInfo	Gets an array of doubles that describe the geometry of the arrowhead on the far end of the leader line.
	IGetAttachPos	Gets the attachment point of the GTol.
	IGetFontInfo	Gets with the font information for this GTol.
	IGetFrameDiameterSymbols	Gets whether diameter symbols are displayed for the specified frame.

		
	IGetFrameSymbols	Obsolete. Superseded by IGtol:IGetFrameSymbols2 .
	IGetFrameSymbols2	Obsolete. Superseded by IGtol:IGetFrameSymbols3 .
	IGetFrameValues	Gets an array that describes the text that appears in each of the fields of the specified GTol frame.
	IGetLeaderAtIndex	Obsolete. Superseded by IGtol:IGetLeaderAtIndex2 .
	IGetLeaderAtIndex2	Gets information about the specified leader on this GTol.
	IGetLeaderInfo	Gets information describing the layout of the GTol leader lines.
	IGetLineAtIndex	Gets the start and end information for the specified line.
	IGetNextGTOL	Gets the next GTol.
	IGetSymArcs	Gets an array that defines all arcs in the symbol.
	IGetSymCircles	Gets an array that defines all circles in the symbol.
	IGetSymEdgeCounts	Gets an array of the number of lines, arcs and circles in the specified symbol.
	IGetSymLines	Gets an array that defines all lines in the symbol.
	IGetSymText	Gets the symbol text.
	IGetSymTextPoints	Gets the text points for the specified symbol.
	IGetTextFormat	Obsolete. Superseded by IAnnotation:IGetTextFormat .
	IGetTextPoint	Gets the GTol text reference point, which is the upper-left corner of the bounding rectangle.
	IGetTextPositionAtIndex	Gets the reference position for the specified text item in this GTol (for example, upper-left, lower-left, center).
	IGetTriangleAtIndex	Gets the triangle at the specified index.
	IGetWitnessLine	Gets the extension line that connects the leader of this geometric tolerance with the entity that is being dimensioned.
	InsertBelowFrameTextAt	Inserts the specified text at the specified line index in the below frame of this GTol.
	IsAttached	Gets whether this GTol is attached.
	ISetTextFormat	Obsolete. Superseded by IAnnotation:ISetTextFormat .
	SetAllAroundThisSide	Sets whether this GTol uses an All Around This Side leader.
	SetAllOver	Sets whether this GTol uses an All Over leader.
	SetAllOverThisSide	Sets whether this GTol uses an All Over This Side leader.
	SetBelowFrameTextAt	Edits or inserts a text line at the specified below frame line index of this GTol.
	SetBetweenTwoPoints	Enables or disables the between two points symbol and its texts.
	SetCompositeFrame	Obsolete. Superseded by IGtol:SetCompositeFrame2 .
	SetCompositeFrame2	Sets whether to create a composite frame containing the specified GTol frame.
	SetDatumIdentifier	Sets the name of the datum being defined.
	SetDisplayDualDimensionInRangeValues	Sets whether to display dual units in dimension range values.
	SetFrameSymbols	Obsolete. Superseded by IGtol:SetFrameSymbols2 .
	SetFrameSymbols2	Sets the symbols for the specified frame.
	SetFrameValues	Obsolete. Superseded by IGtol:SetFrameValues2 .
	SetFrameValues2	Sets the geometric tolerance values and datum references in the specified frame of this GTol symbol.
	SetFromToText	Adds a From-To symbol and sets the From-To text for this GTol.
	SetLeader	Sets the characteristics of the leader for this symbol.
	SetPosition	Sets the position of this GTol.
	SetPTZHeight	Obsolete. Superseded by IGtol:SetPTZHeight2 .
	SetPTZHeight2	Sets the projected tolerance zone for the specified frame and tolerance in this GTol.
	SetText	Sets the specified text part of this GTol.
	SetTextFormat	Obsolete. Superseded by IAnnotation:SetTextFormat .



[Top](#)

See Also

[IGtol Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IAnnotation Interface](#)
[IModelDoc2::InsertGtol Method](#)

SOLIDWORKS API Help

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IGtol Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IGtol Interface

The following tables list the members exposed by [IGtol](#).

Public Properties

Name	Description
 Angle	Gets or sets the rotation angle of this Gtol symbol.
 SeparateRequirement	Gets or sets whether a SEP REQ'T label is present with the Gtol symbol.

[Top](#)

Public Methods

Name	Description
 AddFrame	Adds a frame at the end of the list of this Gtol's frames.
 CanConvertFormat	Gets whether this Gtol can be converted to the current format.
 ConvertFormat	Converts this Gtol to the current format.
 DeleteBelowFrameTextAt	Deletes the specified text line in the below frame of this GTol.
 DeleteFrame	Deletes the specified frame from this Gtol.
 GetAllAround	Gets whether this GTol uses an All Around leader.
 GetAllAroundThisSide	Gets whether this GTol uses an All Around This Side leader.
 GetAllOver	Gets whether this GTol uses an All Over leader.
 GetAllOverThisSide	Gets whether this GTol uses an All Over This Side leader.
 GetAnnotation	Gets the annotation for this specific Gtol.
 GetArcAtIndex	Gets information about the specified arc.
 GetArcCount	Gets the number of arcs in this GTol.
 GetArrowHeadAtIndex	Gets information on the specified arrow head in this GTol.
 GetArrowHeadCount	Gets the number of arrow heads in the GTol.
 GetArrowHeadInfo	Gets an array of doubles that describe the geometry of the arrowhead on the far end of the leader line.
 GetAttachPos	Gets the attachment point of the GTol.
 GetBelowFrameTextAt	Gets the specified line of text in this GTol.
 GetBelowFrameTextLineCount	Gets the number of text lines in the below frame of this GTol.
 GetBetweenTwoPoints	Gets whether the between two points symbol is being used.
 GetBetweenTwoPointsText	Gets the text used in the between two points symbol.
 GetCompositeFrame	Obsolete. Superseded by IGtol::GetCompositeFrame2.
 GetCompositeFrame2	Gets whether the GTol frame with the specified index is part of a composite frame.
 GetDatumIdentifier	Gets the datum identifier.
 GetDisplayDualDimensionInRangeValues	Gets whether to display dual units in dimension range values.
 GetFontInfo	Gets with the font information for this GTol.

		
	GetFormat	Gets the format type of this GTol.
	GetFrame	Gets the GTol frame at the specified index.
	GetFrameCount	Gets the number of frames in this GTol symbol.
	GetFrameDiameterSymbols	Gets whether diameter symbols are displayed for the specified frame.
	GetFrameSymbols	Obsolete. Superseded by IGTol::GetFrameSymbols2 .
	GetFrameSymbols2	Obsolete. Superseded by IGTol::GetFrameSymbols3 .
	GetFrameSymbols3	Gets the symbols for the specified frame.
	GetFrameValues	Gets an array that describes the text that appears in each of the fields of the specified GTol frame.
	GetFromTo	Gets whether the From-To symbol is present in this GTol.
	GetFromToText	Gets the From-To text of this GTol.
	GetHeight	Gets the GTol height.
	GetLeaderAtIndex	Obsolete. Superseded by IGTol::GetLeaderAtIndex2 .
	GetLeaderAtIndex2	Gets information about the specified leader on this GTol.
	GetLeaderCount	Gets the number of leaders on this GTol.
	GetLeaderInfo	Gets information describing the layout of the GTol leader lines.
	GetLeaderSide	Gets where the leader attaches to the symbol.
	GetLineAtIndex	Gets the start and end information for the specified line.
	GetLineCount	Gets the number of linea in this GTol.
	GetNextGTOL	Gets the next GTol.
	GetPTZHeight	Obsolete. Superseded by IGTol::GetPTZHeight2 .
	GetPTZHeight2	Gets the projected tolerance zone for the specified frame and tolerance in this GTol.
	GetSymArcs	Gets an array that defines all arcs in the symbol.
	GetSymCircles	Gets an array that defines all circles in the symbol.
	GetSymDesc	Gets the specified symbol description.
	GetSymEdgeCounts	Gets an array of the number of lines, arcs and circles in the specified symbol.
	GetSymLines	Gets an array that defines all lines in the symbol.
	GetSymName	Gets the symbol name based on the specified index number that SOLIDWORKS uses to identify the symbol.
	GetSymText	Gets the symbol text.
	GetSymTextPoints	Gets the text points for the specified symbol.
	GetText	Gets the specified text part of this GTol.
	GetTextAngleAtIndex	Gets the text angle for the specified piece of text in this GTol.
	GetTextAtIndex	Gets the specified text from this GTol.
	GetTextCount	Gets the number of text items in the GTol.
	GetTextFont	Gets the font for this GTol.
	GetTextFormat	Obsolete. Superseded by IAnnotation::GetTextFormat .
	GetTextHeightAtIndex	Gets the text height for the specified piece of text in this GTol.
	GetTextInvertAtIndex	Gets the specified text item's invert flag.
	GetTextPoint	Gets the GTol text reference point, which is the upper-left corner of the bounding rectangle.
	GetTextPositionAtIndex	Gets the text item's offset relative to note's text point.
	GetTextRefPositionAtIndex	Gets the reference position for the specified text item in this GTol (for example, upper-left, lower-left, center).
	GetTriangleAtIndex	Gets the triangle at the specified index.
	GetTriangleCount	Gets the number of triangles in this GTol.

		
	GetUseDocTextFormat	Obsolete. Superseded by IAnnotation::GetUseDocTextFormat .
	GetWitnessLine	Gets the extension line that connects the leader of this geometric tolerance with the entity being dimensioned.
	HasExtralLeader	Gets whether the GTol has a bent leaderline or a straight leaderline.
	IGetAnnotation	Gets the annotation for this specific GTol.
	IGetArcAtIndex	Gets information for the specified arc.
	IGetArrowHeadAtIndex	Gets information on the specified arrow head in this GTol.
	IGetArrowHeadInfo	Gets an array of doubles that describe the geometry of the arrowhead on the far end of the leader line.
	IGetAttachPos	Gets the attachment point of the GTol.
	IGetFontInfo	Gets with the font information for this GTol.
	IGetFrameDiameterSymbols	Gets whether diameter symbols are displayed for the specified frame.
	IGetFrameSymbols	Obsolete. Superseded by IGtol::IGetFrameSymbols2 .
	IGetFrameSymbols2	Obsolete. Superseded by IGtol::GetFrameSymbols3 .
	IGetFrameValues	Gets an array that describes the text that appears in each of the fields of the specified GTol frame.
	IGetLeaderAtIndex	Obsolete. Superseded by IGtol::IGetLeaderAtIndex2 .
	IGetLeaderAtIndex2	Gets information about the specified leader on this GTol.
	IGetLeaderInfo	Gets information describing the layout of the GTol leader lines.
	IGetLineAtIndex	Gets the start and end information for the specified line.
	IGetNextGTOL	Gets the next GTol.
	IGetSymArcs	Gets an array that defines all arcs in the symbol.
	IGetSymCircles	Gets an array that defines all circles in the symbol.
	IGetSymEdgeCounts	Gets an array of the number of lines, arcs and circles in the specified symbol.
	IGetSymLines	Gets an array that defines all lines in the symbol.
	IGetSymText	Gets the symbol text.
	IGetSymTextPoints	Gets the text points for the specified symbol.
	IGetTextFormat	Obsolete. Superseded by IAnnotation::IGetTextFormat .
	IGetTextPoint	Gets the GTol text reference point, which is the upper-left corner of the bounding rectangle.
	IGetTextPositionAtIndex	Gets the reference position for the specified text item in this GTol (for example, upper-left, lower-left, center).
	IGetTriangleAtIndex	Gets the triangle at the specified index.
	IGetWitnessLine	Gets the extension line that connects the leader of this geometric tolerance with the entity that is being dimensioned.
	InsertBelowFrameTextAt	Inserts the specified text at the specified line index in the below frame of this GTol.
	IsAttached	Gets whether this GTol is attached.
	ISetTextFormat	Obsolete. Superseded by IAnnotation::ISetTextFormat .
	SetAllAroundThisSide	Sets whether this GTol uses an All Around This Side leader.
	SetAllOver	Sets whether this GTol uses an All Over leader.
	SetAllOverThisSide	Sets whether this GTol uses an All Over This Side leader.
	SetBelowFrameTextAt	Edits or inserts a text line at the specified below frame line index of this GTol.
	SetBetweenTwoPoints	Enables or disables the between two points symbol and its texts.
	SetCompositeFrame	Obsolete. Superseded by IGtol::SetCompositeFrame2 .
	SetCompositeFrame2	Sets whether to create a composite frame containing the specified GTol frame.
	SetDatumIdentifier	Sets the name of the datum being defined.
	SetDisplayDualDimensionInRangeValues	Sets whether to display dual units in dimension range values.
	SetFrameSymbols	Obsolete. Superseded by IGtol::SetFrameSymbols2 .

		
	SetFrameSymbols2	Sets the symbols for the specified frame.
	SetFrameValues	Obsolete. Superseded by IGtol.SetFrameValues2 .
	SetFrameValues2	Sets the geometric tolerance values and datum references in the specified frame of this Gtol symbol.
	SetFromToText	Adds a From-To symbol and sets the From-To text for this Gtol.
	SetLeader	Sets the characteristics of the leader for this symbol.
	SetPosition	Sets the position of this Gtol.
	SetPTZHeight	Obsolete. Superseded by IGtol.SetPTZHeight2 .
	SetPTZHeight2	Sets the projected tolerance zone for the specified frame and tolerance in this Gtol.
	SetText	Sets the specified text part of this Gtol.
	SetTextFormat	Obsolete. Superseded by IAnnotation.SetTextFormat .

[Top](#)

 See Also

[IGtol Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IAnnotation Interface](#)
[IModelDoc2::InsertGtol Method](#)

SOLIDWORKS API Help

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IGtol Interface Properties

[See Also](#)

	SolidWorks.Interop.sldworks Namespace : IGtol Interface
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For a list of all members of this type, see [IGtol members](#).

 Public Properties

Name	Description
 Angle	Gets or sets the rotation angle of this Gtol symbol.
 SeparateRequirement	Gets or sets whether a SEP REQ T label is present with the Gtol symbol.

[Top](#)

 See Also

[IGtol Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IAnnotation Interface](#)
[IModelDoc2::InsertGtol Method](#)

SOLIDWORKS API Help

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AddFrame Method (IGtol)

[See Also](#)

		
SolidWorks.Interop.sldworks Namespace	>	IGtol Interface : AddFrame Method (IGtol)

Adds a frame at the end of the list of this Gtol's frames.

 .NET Syntax

Visual Basic (Declaration)

Function AddFrame() As System.Boolean

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.AddFrame()
```

C#

```
System.bool AddFrame()
```

C++/CLI

```
System.bool AddFrame();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if frame successfully added, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::AddFrame](#).

Example

See the [IGtolFrame](#) examples.

Remarks

This method is valid only if this Gtol was created in SOLIDWORKS 2022 or later.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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Angle Property (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : Angle Property (IGtol)

Gets or sets the rotation angle of this Gtol symbol.

.NET Syntax

Visual Basic (Declaration)

```
Property Angle As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Double  
  
instance.Angle = value  
  
value = instance.Angle
```

C#

```
System.double Angle {get; set;}
```

C++/CLI

```
property System.double Angle {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Rotation angle in radians

Visual Basic for Applications (VBA) Syntax

See [Gtol::Angle](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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CanConvertFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : CanConvertFormat Method (IGtol)

Gets whether this Gtol can be converted to the current format.

.NET Syntax

Visual Basic (Declaration)

```
Function CanConvertFormat() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Boolean  
  
value = instance.CanConvertFormat()
```

C#

```
System.bool CanConvertFormat()
```

C++/CLI

```
System.bool CanConvertFormat();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if can be converted, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::CanConvertFormat](#).

Example

See the [IGtolFrame](#) examples.

Remarks

If this method returns true, call [IGtol::ConvertFormat](#) to convert the Gtol to the current format.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

ConvertFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : ConvertFormat Method (IGtol)

Converts this Gtol to the current format.

.NET Syntax

Visual Basic (Declaration)

```
Function ConvertFormat() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Integer

value = instance.ConvertFormat()
```

C#

```
System.int ConvertFormat()
```

C++/CLI

```
System.int ConvertFormat();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

0 if successfully converted, otherwise error code as defined by [swGtolFormatConversionError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Gtol::ConvertFormat](#).

Example

See the [IGtolFrame](#) examples.

Remarks

Before calling this method, use [IGtol::CanConvertFormat](#) to determine whether conversion is possible.
If unsupported data are in this older Gtol, then this method will fail.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

DeleteBelowFrameTextAt Method (IGTol)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IGTol Interface](#) : DeleteBelowFrameTextAt Method (IGTol)

Deletes the specified text line in the below frame of this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteBelowFrameTextAt( _
    ByVal Index As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IGTol
Dim Index As System.Integer
Dim value As System.Boolean

value = instance.DeleteBelowFrameTextAt(Index)
```

C#

```
System.bool DeleteBelowFrameTextAt(
    System.int Index
)
```

C++/CLI

```
System.bool DeleteBelowFrameTextAt(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

1-based index of the below frame text line to delete; -1 to delete all of the below frame text lines

Return Value

True if text line successfully deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::DeleteBelowFrameTextAt](#).

Example

See **Set Text in Datum Tags and GTols** examples in [IGTol](#).

See Also

[IGTol Interface](#)
[IGTol Members](#)
[IGtol::GetBelowFrameTextAt Method \(\)](#)
[IGtol::GetBelowFrameTextLineCount Method \(\)](#)
[IGtol::InsertBelowFrameTextAt Method \(\)](#)
[IGtol::SetBelowFrameTextAt Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : DeleteFrame Method (IGtol)

Deletes the specified frame from this Gtol.

.NET Syntax

Visual Basic (Declaration)

```
Function DeleteFrame( _  
    ByVal FrameNum As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNum As System.Integer  
Dim value As System.Boolean  
  
value = instance.DeleteFrame(FrameNum)
```

C#

```
System.bool DeleteFrame(  
    System.int FrameNum  
)
```

C++/CLI

```
System.bool DeleteFrame(  
& System.int FrameNum  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNum
One-based index of Gtol to delete

Return Value

True if successfully deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol.DeleteFrame](#).

Example

See the [IGtolFrame](#) examples.

Remarks

This method is valid only if this Gtol was created in SOLIDWORKS 2022 or later.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetAllAround Method (IGtol)

Gets whether this GTol uses an All Around leader.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAllAround() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.GetAllAround()
```

C#

```
System.bool GetAllAround()
```

C++/CLI

```
System.bool GetAllAround();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this GTol uses an All Around leader, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetAllAround](#).

Remarks

This property is valid only for bent, perpendicular, and multi-jog leaders. Call [IGtol::GetLeaderAtIndex2](#) to determine which type of leader is set for this GTol.

Use:

- [IGtol::IsAttached](#) to determine whether this symbol is currently using a leader.
- [IGtol::HasExtraLeader](#) to determine whether this symbol is using a bent leader.
- [IGtol::GetLeaderSide](#) to determine where the leader is attached to the symbol.
- [IGtol::SetLeader](#) to set the characteristics of the leader on this symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetAllAroundThisSide Method \(\)](#)
[IGtol::GetAllOverThisSide Method \(\)](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAllAroundThisSide Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetAllAroundThisSide Method (IGtol)

Gets whether this GTol uses an All Around This Side leader.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAllAroundThisSide() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Boolean
```

```
value = instance.GetAllAroundThisSide()
```

C#

```
System.bool GetAllAroundThisSide()
```

C++/CLI

```
System.bool GetAllAroundThisSide();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this GTol uses an All Around This Side leader, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetAllAroundThisSide](#).

Remarks

This property is valid only for bent, perpendicular, and multi-jog leaders. Call [IGtol::GetLeaderAtIndex2](#) to determine which type of leader is set for this GTol.

Use:

- [IGtol::IsAttached](#) to determine whether this symbol is currently using a leader.
- [IGtol::HasExtraLeader](#) to determine whether this symbol is using a bent leader.
- [IGtol::GetLeaderSide](#) to determine where the leader is attached to the symbol.
- [IGtol::SetLeader](#) to set the characteristics of the leader on this symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetAllAroundThisSide Method \(\)](#)
[IGtol::GetAllOverThisSide Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAllOver Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetAllOver Method (IGtol)

Gets whether this GTol uses an All Over leader.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAllOver() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.GetAllOver()
```

C#

```
System.bool GetAllOver()
```

C++/CLI

```
System.bool GetAllOver();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this GTol uses an All Over leader, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetAllOver](#).

Remarks

This property is valid only for bent, perpendicular, and multi-jog leaders. Call [IGtol::GetLeaderAtIndex2](#) to determine which type of leader is set for this GTol.

Use:

- [IGtol::IsAttached](#) to determine whether this symbol is currently using a leader.
- [IGtol::HasExtraLeader](#) to determine whether this symbol is using a bent leader.
- [IGtol::GetLeaderSide](#) to determine where the leader is attached to the symbol.
- [IGtol::SetLeader](#) to set the characteristics of the leader on this symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetAllOver Method \(\)](#)
[IGtol::GetAllAround Method \(\)](#)
[IGtol::GetAllAroundThisSide Method \(\)](#)
[IGtol::GetAllOverThisSide Method \(\)](#)

Availability

SOLIDWORKS 2026 SP2, Revision Number 34.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAllOverThisSide Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetAllOverThisSide Method (IGtol)

Gets whether this GTol uses an All Over This Side leader.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAllOverThisSide() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.GetAllOverThisSide()
```

C#

```
System.bool GetAllOverThisSide()
```

C++/CLI

```
System.bool GetAllOverThisSide();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this GTol uses an All Over This Side leader, false if not

Visual Basic for Applications (VBA) Syntax

See [GTol::GetAllOverThisSide](#).

Remarks

This property is valid only for bent, perpendicular, and multi-jog leaders. Call [IGtol::GetLeaderAtIndex2](#) to determine which type of leader is set for this GTol.

Use:

- [IGtol::IsAttached](#) to determine whether this symbol is currently using a leader.
- [IGtol::HasExtraLeader](#) to determine whether this symbol is using a bent leader.
- [IGtol::GetLeaderSide](#) to determine where the leader is attached to the symbol.
- [IGtol::SetLeader](#) to set the characteristics of the leader on this symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetAllOverThisSide Method \(\)](#)
[IGtol::GetAllAroundThisSide Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAnnotation Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetAnnotation Method (IGtol)

Gets the annotation for this specific GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAnnotation() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Object  
  
value = instance.GetAnnotation()
```

C#

```
System.object GetAnnotation()
```

C++/CLI

```
System.Object^ GetAnnotation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IAnnotation](#) object

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetAnnotation](#).

Example

[Get Annotation Object \(VBA\)](#)
[Get Annotation Object \(VB.NET\)](#)
[Get Annotation Object \(C#\)](#)

Remarks

The IAnnotation object is a higher-level object that contains methods that are available to all types of annotations.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::IGetAnnotation Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetArcAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetArcAtIndex Method (IGtol)

Gets information about the specified arc.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetArcAtIndex(Index)
```

C#

```
System.object GetArcAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetArcAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the arc

Return Value

Array of doubles (see **Remarks**)

 Visual Basic for Applications (VBA) Syntax

See [Gtol::GetArcAtIndex](#).

 Remarks

The return value is the following array of doubles:

[*lineType*, *startPt*[3], *endPt*[3], *centerPt*[3], *rotationDir*]

where:

<i>lineType</i>	Line type
<i>startPt</i> [3]	XYZ arc start point
<i>endPt</i> [3]	XYZ arc end point
<i>centerPt</i> [3]	XYZ arc center point
<i>rotationDir</i>	Boolean returned as a double and represents the rotation direction, where CCW returns True and CW returns false

 See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::IGetArcAtIndex Method](#)
[IGtol::GetArcCount Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetArcCount Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetArcCount Method (IGtol)

Gets the number of arcs in this GTol.

 .NET Syntax

Visual Basic (Declaration)

Function GetArcCount() As System.Integer

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Integer  
  
value = instance.GetArcCount()
```

C#

```
System.int GetArcCount()
```

C++/CLI

```
System.int GetArcCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of arcs

 Visual Basic for Applications (VBA) Syntax

See [Gtol::GetArcCount](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetArcAtIndex Method](#)
[IGtol::IGetArcAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetArrowHeadAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetArrowHeadAtIndex Method (IGtol)

Gets information on the specified arrow head in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArrowHeadAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetArrowHeadAtIndex(Index)
```

C#

```
System.Object GetArrowHeadAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetArrowHeadAtIndex(  
    & System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the arrow head

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetArrowHeadAtIndex](#).

Remarks

The return value is an array of doubles in the following format:

[*arrowHeadPt*[3], *arrowHeadDir*[3], *arrowHeadWidth*, *arrowHeadHeight*, *arrowHeadStyle*]

where:

<i>arrowHeadPt</i> [3]	XYZ arrow head tip location
<i>arrowHeadDir</i> [3]	XYZ arrow head direction
<i>arrowHeadWidth</i>	Arrow head width where the width is measured along the arrow head direction

<i>arrowHeadHeight</i>	Arrow head height
<i>arrowHeadStyle</i>	Arrow head style as defined in swArrowStyle_e

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::IGetArrowHeadAtIndex Method](#)
[IGtol::IGetArrowHeadCount Method](#)
[IGtol::IGetArrowHeadInfo Method](#)
[IGtol::IGetArrowHeadInfo Method](#)

SOLIDWORKS API Help

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GetArrowHeadCount Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetArrowHeadCount Method (IGtol)

Gets the number of arrow heads in the GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArrowHeadCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Integer

value = instance.GetArrowHeadCount()
```

C#

```
System.int GetArrowHeadCount()
```

C++/CLI

```
System.int GetArrowHeadCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of arrow heads

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetArrowHeadCount](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetArrowHeadAtIndex Method](#)
[IGtol::GetArrowHeadInfo Method](#)
[IGtol::IGetArrowHeadAtIndex Method](#)
[IGtol::IGetArrowHeadInfo Method](#)

SOLIDWORKS API Help

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GetArrowHeadInfo Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetArrowHeadInfo Method (IGtol)

Gets an array of doubles that describe the geometry of the arrowhead on the far end of the leader line.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArrowHeadInfo() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Object  
  
value = instance.GetArrowHeadInfo()
```

C#

```
System.object GetArrowHeadInfo()
```

C++/CLI

```
System.Object^ GetArrowHeadInfo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetArrowHeadInfo](#).

Remarks

This information returned by this method is independent of whether this note actually has an arrowhead.

The format of the return value is the following array of doubles:

retval[0] = Arrow length (for example, leader into arrowhead)
retval[1] = Arrowhead length
retval[2] = Arrowhead width
retval[3] = Arrowhead style as defined in [swArrowStyle_e](#)

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetArrowHeadAtIndex Method](#)
[IGtol::GetArrowHeadCount Method](#)
[IGtol::IGetArrowHeadAtIndex Method](#)
[IGtol::IGetArrowHeadInfo Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetAttachPos Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetAttachPos Method (IGtol)

Gets the attachment point of the GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAttachPos() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Object  
  
value = instance.GetAttachPos()
```

C#

```
System.Object GetAttachPos()
```

C++/CLI

```
System.Object^ GetAttachPos();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetAttachPos](#).

Remarks

This method is meaningful only if this GTol is attached.

Format of return value is the following array of doubles:

retval[0] = X-coordinate of attachment point

retval[1] = Y-coordinate of attachment point

retval[2] = Z-coordinate of attachment point

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::IGetAttachPos Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBelowFrameTextAt Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetBelowFrameTextAt Method (IGtol)

Gets the specified line of text in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBelowFrameTextAt( _  
    ByVal Index As System.Integer _  
) As System.String
```

Visual Basic (Usage)

```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.String  
  
value = instance.GetBelowFrameTextAt(Index)
```

C#

```
System.string GetBelowFrameTextAt(  
    System.int Index  
)
```

C++/CLI

```
System.String^ GetBelowFrameTextAt(  
& System.int Index  
)
```



NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

1-based index of the line whose text to get

Return Value

Below frame line of text

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetBelowFrameTextAt](#).

Example

See **Set Text in Datum Tags and GTols** examples in [IGTol](#).

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::DeleteBelowFrameTextAt Method \(\)](#)

[IGtol::GetBelowFrameTextLineCount Method \(\)](#)

[IGtol::InsertBelowFrameTextAt Method \(\)](#)

[IGtol::SetBelowFrameTextAt Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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GetBelowFrameTextLineCount Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetBelowFrameTextLineCount Method (IGtol)

Gets the number of text lines in the below frame of this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBelowFrameTextLineCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol
```

```
Dim value As System.Integer
```

```
value = instance.GetBelowFrameTextLineCount()
```

C#

```
System.int GetBelowFrameTextLineCount()
```

C++/CLI

```
System.int GetBelowFrameTextLineCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of text lines in the below frame

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetBelowFrameTextLineCount](#).

Example

See **Set Text in Datum Tags and GTols** examples in [IGTol](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::DeleteBelowFrameTextAt Method \(\)](#)
[IGtol::GetBelowFrameTextAt Method \(\)](#)
[IGtol::InsertBelowFrameTextAt Method \(\)](#)
[IGtol::SetBelowFrameTextAt Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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GetBetweenTwoPoints Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetBetweenTwoPoints Method (IGtol)

Gets whether the between two points symbol is being used.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBetweenTwoPoints() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Boolean  
  
value = instance.GetBetweenTwoPoints()
```

C#

```
System.bool GetBetweenTwoPoints()
```

C++/CLI

```
System.bool GetBetweenTwoPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if using the between two points symbol, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetBetweenTwoPoints](#).

Remarks

Use

- [IGtol::GetBetweenTwoPointsText](#) to get the text that is used with this symbol.
- [IGtol::SetBetweenTwoPoints](#) to enable this symbol and its texts.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetBetweenTwoPointsText Method](#)
[IGtol::SetBetweenTwoPoints Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

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GetBetweenTwoPointsText Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetBetweenTwoPointsText Method (IGtol)

Gets the text used in the between two points symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBetweenTwoPointsText( _  
    ByVal Index As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.String  
  
value = instance.GetBetweenTwoPointsText(Index)
```

C#

```
System.string GetBetweenTwoPointsText(  
    System.int Index  
)
```

C++/CLI

```
System.String^ GetBetweenTwoPointsText(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0 for the text on the left end of the symbol, 1 for the text on the right end of the symbol

Return Value

Symbol text

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetBetweenTwoPointsText](#).

Remarks

This method returns the requested text whether the between two points symbol is currently enabled. Use [IGtol::GetBetweenTwoPoints](#) to determine if this symbol is enabled.

Use [IGtol::SetBetweenTwoPoints](#) to enable this symbol and its texts.

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::SetBetweenTwoPoints Method](#)

[IGtol::GetBetweenTwoPoints Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCompositeFrame Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetCompositeFrame Method (IGtol)

Obsolete. Superseded by [IGtol::GetCompositeFrame2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetCompositeFrame() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.GetCompositeFrame()
```

C#

```
System.bool GetCompositeFrame()
```

C++/CLI

```
System.bool GetCompositeFrame();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the first two frames of this Gtol are in a composite frame, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetCompositeFrame](#).

Remarks

Gets whether the first two frames of this Gtol are in a composite frame. Use [IGtol::SetCompositeFrame](#) to set this value.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 998Plus, datecode 1998319

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCompositeFrame2 Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetCompositeFrame2 Method (IGtol)

Gets whether the Gtol frame with the specified index is part of a composite frame.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCompositeFrame2( _  
    ByVal FrameNum As System.Short _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNum As System.Short  
Dim value As System.Boolean  
  
value = instance.GetCompositeFrame2(FrameNum)
```

C#

```
System.bool GetCompositeFrame2(  
    System.short FrameNum  
)
```

C++/CLI

```
System.bool GetCompositeFrame2(  
& System.short FrameNum  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNum
Index of GTol frame

Return Value

True if GTol frame is part of a composite frame, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetCompositeFrame2](#).

Example

[Create GTol Composite Frame \(VBA\)](#)
[Create GTol Composite Frame \(VB.NET\)](#)
[Create GTol Composite Frame \(C#\)](#)

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetCompositeFrame2 Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetDatumIdentifier Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetDatumIdentifier Method (IGtol)

Gets the datum identifier.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDatumIdentifier() As System.String
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.String

value = instance.GetDatumIdentifier()
```

C#

```
System.string GetDatumIdentifier()
```

C++/CLI

```
System.String^ GetDatumIdentifier();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Datum identifier

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetDatumIdentifier](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetDatumIdentifier Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDisplayDualDimensionInRangeValues Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetDisplayDualDimensionInRangeValues Method (IGtol)

Gets whether to display dual units in dimension range values.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDisplayDualDimensionInRangeValues() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.GetDisplayDualDimensionInRangeValues()
```

C#

```
System.bool GetDisplayDualDimensionInRangeValues()
```

C++/CLI

```
System.bool GetDisplayDualDimensionInRangeValues();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True to display dual units in dimension range values, false to not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetDisplayDualDimensionInRangeValues](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetDisplayDualDimensionInRangeValues Method \(\)](#)

 Availability

SOLIDWORKS 2024 FCS, Revision Number 32

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFontInfo Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFontInfo Method (IGtol)

Gets with the font information for this GTol.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetFontInfo() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Object

value = instance.GetFontInfo()
```

C#

```
System.object GetFontInfo()
```

C++/CLI

```
System.Object^ GetFontInfo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object containing the font information (see **Remarks**)

 Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFontInfo](#).

 Remarks

Format of return value is an array of doubles. Currently this consists only of a [line type index](#).

 See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::IGetFontInfo Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFormat Method (IGtol)

Gets the format type of this GTol.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetFormat() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Integer

value = instance.GetFormat()
```

C#

```
System.int GetFormat()
```

C++/CLI

```
System.int GetFormat();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of Gtol format as defined by [swGtolFormatType_e](#)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFormat](#).

Example

See the [IGtolFrame](#) examples.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFrame Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFrame Method (IGtol)

Gets the Gtol frame at the specified index.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFrame( _
    ByVal FrameIndex As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim FrameIndex As System.Integer
Dim value As System.Object

value = instance.GetFrame(FrameIndex)
```

C#

```
System.Object GetFrame(  
    System.int FrameIndex  
)
```

C++/CLI

```
System.Object^ GetFrame(  
& System.int FrameIndex  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameIndex
One-based index of Gtol frame to retrieve

Return Value

[IGtolFrame](#)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFrame](#).

Example

See the [IGtolFrame](#) examples.

Remarks

This method is valid only if this Gtol was created in SOLIDWORKS 2022 or later.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetFrameCount Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFrameCount Method (IGtol)

Gets the number of frames in this Gtol symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFrameCount() As System.Short
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Short  
  
value = instance.GetFrameCount()
```

C#

```
System.short GetFrameCount()
```

C++/CLI

```
System.Short GetFrameCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of frames in this GTol symbol

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFrameCount](#).

Example

See the [IGtolFrame](#) examples.

Example

[Get Text Items in GTol Frame \(VBA\)](#)
[Get Text Items in GTol Frame \(VB.NET\)](#)
[Get Text Items in GTol Frame \(C#\)](#)

Remarks

This method returns the number of frames that this symbol has stored, which may be different than the number of frames that actually appear when this frame is displayed. This is because symbols can be created, either through the user interface or the API, with empty frames or rows, which do not appear when the frame is displayed.

For example, the symbol can be displayed with three 3 frames, but upon examination, it could be that those are frames 1, 2, and 5. Frames 3 and 4 are empty. Every symbol must have at least 2 frames.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameDiameterSymbols Method](#)
[IGtol::GetFrameSymbols2 Method](#)
[IGtol::GetFrameValues Method](#)
[IGtol::IGetFrameDiameterSymbols Method](#)
[IGtol::IGetFrameSymbols2 Method](#)
[IGtol::IGetFrameValues Method](#)
[IGtol::SetFrameSymbols2 Method](#)
[IGtol::SetFrameValues Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFrameDiameterSymbols Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFrameDiameterSymbols Method (IGtol)

Gets whether diameter symbols are displayed for the specified frame.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFrameDiameterSymbols( _  
    ByVal FrameNumber As System.Short _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IGtol  
Dim FrameNumber As System.Short  
Dim value As System.Object
```



```
value = instance.GetFrameDiameterSymbols(FrameNumber)
```

C#

```
System.Object GetFrameDiameterSymbols(  
    System.short FrameNumber  
)
```

C++/CLI

```
System.Object^ GetFrameDiameterSymbols(  
& System.short FrameNumber  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Frame number to examine (1 or 2)

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFrameDiameterSymbols](#).

Remarks

This method is valid only if this Gtol was created in a version of SOLIDWORKS earlier than 2022.

The return value should be an array of two Booleans, with the following format:

retval[0] = True if the first tolerance value has a diameter symbol, false if it does not

retval[1] = True if the second tolerance value has a diameter symbol, false if it does not

Use this method in combination with [IGtol::GetFrameSymbols2](#) that is used to retrieve the frame symbols.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameCount Method](#)
[IGtol::GetFrameSymbols2 Method](#)
[IGtol::GetFrameValues Method](#)
[IGtol::IGetFrameSymbols2 Method](#)
[IGtol::IGetFrameValues Method](#)
[IGtol::SetFrameSymbols2 Method](#)
[IGtol::SetFrameValues Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFrameSymbols Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFrameSymbols Method (IGtol)

Obsolete. Superseded by [IGtol::GetFrameSymbols2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetFrameSymbols( _  
    ByVal FrameNumber As System.Short _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IGtol
Dim FrameNumber As System.Short
Dim value As System.Object

value = instance.GetFrameSymbols(FrameNumber)
```

C#

```
System.Object GetFrameSymbols(
    System.Short FrameNumber
)
```

C++/CLI

```
System.Object^ GetFrameSymbols(
    & System.Short FrameNumber
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFrameSymbols](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFrameSymbols2 Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFrameSymbols2 Method (IGtol)

Obsolete. Superseded by [IGtol::GetFrameSymbols3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetFrameSymbols2( _
    ByVal FrameNumber As System.Short _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IGtol
Dim FrameNumber As System.Short
Dim value As System.Object

value = instance.GetFrameSymbols2(FrameNumber)
```

C#

```
System.Object GetFrameSymbols2(
    System.Short FrameNumber
)
```

C++/CLI

```
System.Object^ GetFrameSymbols2(
    & System.Short FrameNumber
)
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber
Frame number to examine (1 or 2)

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [IGtol::GetFrameSymbols2](#).

Remarks

The return array is an array of six strings in the following format:

retval[0] = Geometric tolerance symbol
retval[1] = Material condition symbol for first tolerance value
retval[2] = Material condition symbol for second tolerance value
retval[3] = Material condition symbol for datum1
retval[4] = Material condition symbol for datum2
retval[5] = Material condition symbol for datum3

The character strings returned in the array correspond to symbols defined in **C:\ProgramData\SolidWorks\SolidWorks 20nn\languish\gtol.sym**. The format of each string is <LibraryName-SymbolName> (for example, <GTOL-ANGULAR>, which is the angularity symbol from the ASME Geometric Tolerancing Symbols library).

Use this method with [IGtol::GetFrameDiameterSymbols](#), which determines whether diameter symbols are displayed.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameCount Method](#)
[IGtol::GetFrameDiameterSymbols Method](#)
[IGtol::GetFrameValues Method](#)
[IGtol::IGetFrameDiameterSymbols Method](#)
[IGtol::IGetFrameSymbols2 Method](#)
[IGtol::IGetFrameValues Method](#)
[IGtol::SetFrameSymbols2 Method](#)
[IGtol::SetFrameValues Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFrameSymbols3 Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFrameSymbols3 Method (IGtol)

Gets the symbols for the specified frame.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFrameSymbols3( _  
    ByVal FrameNumber As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IGtol  
Dim FrameNumber As System.Integer  
Dim value As System.Object
```



```
value = instance.GetFrameSymbols3(FrameNumber)
```

C#

```
System.object GetFrameSymbols3(  
    System.int FrameNumber  
)
```

C++/CLI

```
System.Object^ GetFrameSymbols3(  
& System.int FrameNumber  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber
Frame number to examine (1 or 2)

Return Value

Array of six strings (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFrameSymbols3](#).

Example

[Get Text Items in GTol Frame \(C#\)](#)
[Get Text Items in GTol Frame \(VB.NET\)](#)
[Get Text Items in GTol Frame \(VBA\)](#)

Remarks

This method is valid only if this Gtol was created in a version of SOLIDWORKS earlier than 2022.

The return array is an array of six strings in the following format:

retval[0] = Geometric tolerance symbol
retval[1] = Material condition symbol for first tolerance value
retval[2] = Material condition symbol for second tolerance value
retval[3] = Material condition symbol for datum1
retval[4] = Material condition symbol for datum2
retval[5] = Material condition symbol for datum3

The character strings returned in the array correspond to symbols defined in **C:\ProgramData\SolidWorks\SolidWorks 20nn\language\gtol.sym**. The format of each string is *<LibraryName-SymbolName>* (for example, **<GTOL-ANGULAR>**, which is the angularity symbol from the ASME Geometric Tolerancing Symbols library).

Use this method with [IGtol::GetFrameDiameterSymbols](#), which determines whether diameter symbols are displayed.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameCount Method \(\)](#)
[IGtol::GetFrameDiameterSymbols Method \(\)](#)
[IGtol::IGetFrameDiameterSymbols Method \(\)](#)
[IGtol::GetFrameValues Method \(\)](#)
[IGtol::IGetFrameValues Method \(\)](#)
[IGtol::SetFrameSymbols2 Method \(\)](#)
[IGtol::SetFrameValues2 Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFrameValues Method (IGtol)

Gets an array that describes the text that appears in each of the fields of the specified GTol frame.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFrameValues( _  
    ByVal FrameNumber As System.Short _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNumber As System.Short  
Dim value As System.Object  
  
value = instance.GetFrameValues(FrameNumber)
```

C#

```
System.object GetFrameValues(  
    System.short FrameNumber  
)
```

C++/CLI

```
System.Object^ GetFrameValues(  
& System.short FrameNumber  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Frame number to examine (1 or 2)

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFrameValues](#).

Example

[Get Text Items in GTol Frame \(VBA\)](#)
[Get Text Items in GTol Frame \(VB.NET\)](#)
[Get Text Items in GTol Frame \(C#\)](#)

Remarks

This method is valid only if this GTol was created in a version of SOLIDWORKS earlier than 2022.

Format of return array of strings is:

retval[0] = Tolerance 1

retval[1] = Tolerance 2

retval[2] = Datum 1

retval[3] = Datum 2

retval[4] = Datum 3

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameCount Method](#)
[IGtol::GetFrameDiameterSymbols Method](#)
[IGtol::GetFrameSymbols2 Method](#)
[IGtol::IGetFrameDiameterSymbols Method](#)
[IGtol::IGetFrameSymbols2 Method](#)
[IGtol::IGetFrameValues Method](#)
[IGtol::SetFrameSymbols2 Method](#)
[IGtol::SetFrameValues Method](#)

SOLIDWORKS API Help

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GetFromTo Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFromTo Method (IGtol)

Gets whether the From-To symbol is present in this Gtol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFromTo() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Boolean

value = instance.GetFromTo()
```

C#

```
System.bool GetFromTo()
```

C++/CLI

```
System.bool GetFromTo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if From-To symbol is present, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFromTo](#).

Example

See the [IGtolFrame](#) examples.

Remarks

The From-To symbol:

- is present below the Gtol frame.
- specifies that the Gtol value applies from one point or entity to another.

If this method returns true, then call [IGtol::GetFromToText](#) and [IGtol::SetFromToText](#) to retrieve and set the From-To labels for this Gtol.

See Also

[IGtol Interface](#)
[IGtol Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetFromToText Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetFromToText Method (IGtol)

Gets the From-To text of this Gtol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFromToText( _  
    ByRef FromText As System.String, _  
    ByRef ToText As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FromText As System.String  
Dim ToText As System.String  
Dim value As System.Boolean  
  
value = instance.GetFromToText(FromText, ToText)
```

C#

```
System.bool GetFromToText(  
    out System.string FromText,  
    out System.string ToText  
)
```

C++/CLI

```
System.bool GetFromToText(  
& [Out] System.String^ FromText,  
& [Out] System.String^ ToText  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FromText
From text label
ToText
To text label

Return Value

True if From-To text successfully retrieved, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetFromToText](#).

Example

See the [IGtolFrame](#) examples.

Remarks

The From-To symbol:

- is present below the Gtol frame.
- specifies that the Gtol value applies from one point or entity to another.

Before calling this method, use [IGtol::GetFromTo](#) to determine whether the From-To symbol is present.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetFromToText Method \(\)](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetHeight Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetHeight Method (IGtol)

Gets the GTol height.

.NET Syntax

Visual Basic (Declaration)

```
Function GetHeight() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Double  
  
value = instance.GetHeight()
```

C#

```
System.double GetHeight()
```

C++/CLI

```
System.double GetHeight();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

GTol height

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetHeight](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLeaderAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetLeaderAtIndex Method (IGtol)

Obsolete. Superseded by [IGtol::GetLeaderAtIndex2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetLeaderAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim Index As System.Integer
Dim value As System.Object

value = instance.GetLeaderAtIndex(Index)
```

C#

```
System.Object GetLeaderAtIndex(
    System.int Index
)
```

C++/CLI

```
System.Object^ GetLeaderAtIndex(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetLeaderAtIndex](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLeaderAtIndex2 Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetLeaderAtIndex2 Method (IGtol)

Gets information about the specified leader on this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLeaderAtIndex2( _
    ByVal Index As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim Index As System.Integer
Dim value As System.Object

value = instance.GetLeaderAtIndex2(Index)
```

C#

```
System.Object GetLeaderAtIndex2(
    System.int Index
)
```

C++/CLI

```
System.Object^ GetLeaderAtIndex2(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the leader

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetLeaderAtIndex2](#).

Remarks

A Gtol can have a straight leader (1 segment) or a bent leader (2 segments), or it might have a perpendicular leader, which can have either 2 or 3 segments, depending on the situation. The returned array contains the point information to define the leader. It contains the x,y,z coordinates for each vertex in the leader. It can contain up to 12 doubles (4 sets of x,y,z values).

Examine the return value to determine the number of items in the leader coordinate information. The size of the array is either 6 (1 segments), 9 (2 segments), or 12 (3 segments).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLeaderCount Method](#)
[IGtol::GetLeaderInfo Method](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::GetLeaderInfo Method](#)
[IGtol::SetLeader Method](#)
[IGtol::GetLeaderSide Method](#)
[IGtol::IGetLineAtIndex Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLeaderCount Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetLeaderCount Method (IGtol)

Gets the number of leaders on this Gtol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLeaderCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Integer

value = instance.GetLeaderCount()
```

C#

```
System.int GetLeaderCount()
```

C++/CLI

```
System.int GetLeaderCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of leaders on the Gtol

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetLeaderCount](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::GetLeaderInfo Method](#)
[IGtol::GetLeaderSide Method](#)
[IGtol::IGetLeaderInfo Method](#)
[IGtol::IGetLeaderInfo Method](#)
[IGtol::SetLeader Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLeaderInfo Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetLeaderInfo Method (IGtol)

Gets information describing the layout of the GTol leader lines.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLeaderInfo() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Object

value = instance.GetLeaderInfo()
```

C#

```
System.object GetLeaderInfo()
```

C++/CLI

```
System.Object^ GetLeaderInfo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetLeaderInfo](#).

Remarks

There can be 0, 1 or 2 lines with the leader line. If the GTol is not attached, then there are 0 lines. Otherwise, the GTol can have a straight leaderline (1 line) or a bent leaderline (2 lines). The caller must infer the number of leader lines based on [IGtol::IsAttached](#) and [IGtol::HasExtraLeader](#).

- IGtol::IsAttached() == false implies no leaderline; therefore, there are no leaderline points (PointCount=0).
- IGtol::HasExtraLeader() == false means that this is a straight leaderline; therefore, there is 1 line (PointCount=2).
- IGtol::HasExtraLeader() == True means that this is a bent leaderline; therefore, there are 2 lines (PointCount=3).

The format of return information is the following array of doubles:

retval[0] = X-coordinate of first leader point

retval[1] = Y-coordinate of first leader point

retval[2] = Z-coordinate of first leader point

retval[3] = X-coordinate of second leader point
retval[4] = Y-coordinate of second leader point
retval[5] = Z-coordinate of second leader point

retval[6] = X-coordinate of third leader point
retval[7] = Y-coordinate of third leader point
retval[8] = Z-coordinate of third leader point

Use [IGtol::GetLeaderCount](#) to see how many leaders are on the [IGTol](#) object.

 See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::GetLeaderSide Method](#)
[IGtol::IGetLeaderAtIndex2 Method](#)
[IGtol::IGetLeaderInfo Method](#)
[IGtol::SetLeader Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLeaderSide Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetLeaderSide Method (IGtol)

Gets where the leader attaches to the symbol.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetLeaderSide() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim value As System.Integer  
  
value = instance.GetLeaderSide()
```

C#

```
System.int GetLeaderSide()
```

C++/CLI

```
System.int GetLeaderSide();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Leader attachment information as defined in [swLeaderSide_e](#)

 Visual Basic for Applications (VBA) Syntax

See [Gtol::GetLeaderSide](#).

 Remarks

Use...	To...
IGtol::IsAttached	Determine if this symbol is using a leader
IGtol::HasExtraLeader	Determine if this symbol is using a bent leader
IGtol::GetAllAround	Determine if this symbol is using an all around symbol

[IGtol::SetLeader](#)

Set the characteristics of the leader on this symbol

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLeaderCount Method](#)
[IGtol::GetLeaderInfo Method](#)
[IGtol::GetLeaderAtIndex2 Method](#)
[IGtol::IGetLeaderAtIndex2 Method](#)
[IGtol::IGetLeaderInfo Method](#)

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GetLineAtIndex Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetLineAtIndex Method (IGtol)

Gets the start and end information for the specified line.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLineAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Index As System.Integer  
Dim value As System.Object  
  
value = instance.GetLineAtIndex(Index)
```

C#

```
System.object GetLineAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.Object^ GetLineAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the desired line where the index begins at zero

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetLineAtIndex](#).

Remarks

The return value is the following array of doubles:

[*lineType*, *startPt*[3], *endPt*[3]]

where:

- *lineType* = the [line type](#)

- *startPt*[3] = the XYZ line start point
- *endPt*[3] = the XYZ line end point

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLineCount Method](#)
[IGtol::IGetLineAtIndex Method](#)

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GetLineCount Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetLineCount Method (IGtol)

Gets the number of linea in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLineCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim value As System.Integer

value = instance.GetLineCount()
```

C#

```
System.int GetLineCount()
```

C++/CLI

```
System.int GetLineCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of lines

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetLineCount](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetLineAtIndex Method](#)
[IGtol::IGetLineAtIndex Method](#)

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GetNextGTOL Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetNextGTOL Method (IGtol)

Gets the next GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetNextGTOL() As System.Object
```

Visual Basic (Usage)

```
Dim instance As IGtol
Dim value As System.Object

value = instance.GetNextGTOL()
```

C#

```
System.object GetNextGTOL()
```

C++/CLI

```
System.Object^ GetNextGTOL();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the next [GTol](#)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetNextGTOL](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::IGetNextGTOL Method](#)
[IAnnotation::GetSpecificAnnotation Method](#)
[IAnnotation::IGetSpecificAnnotation Method](#)
[IDrawingDoc::INewGtol Method](#)
[IDrawingDoc::NewGtol Method](#)
[IView::GetFirstGTOL Method](#)
[IView::IGetFirstGTOL Method](#)

SOLIDWORKS API Help

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GetPTZHeight Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetPTZHeight Method (IGtol)

Obsolete. Superseded by [IGtol::GetPTZHeight2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function GetPTZHeight() As System.String
```

Visual Basic (Usage)

```
Dim instance As IGtol
Dim value As System.String

value = instance.GetPTZHeight()
```

C#

```
System.string GetPTZHeight()
```

C++/CLI

```
System.String^ GetPTZHeight();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Height of the projected tolerance zone (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetPTZHeight](#).

Remarks

This method returns the height of the projected tolerance zone as a string because it is a user-specified parameter that can be textual, numeric, or both types of data.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetPTZHeight Method](#)

SOLIDWORKS API Help

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GetPTZHeight2 Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : GetPTZHeight2 Method (IGtol)

Gets the projected tolerance zone for the specified frame and tolerance in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPTZHeight2( _  
    ByVal FrameNumber As System.Integer, _  
    ByVal TolNumber As System.Integer, _  
    ByRef PtzDisplay As System.Boolean, _  
    ByRef PtzHt As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNumber As System.Integer  
Dim TolNumber As System.Integer  
Dim PtzDisplay As System.Boolean  
Dim PtzHt As System.String  
Dim value As System.Boolean  
  
value = instance.GetPTZHeight2(FrameNumber, TolNumber, PtzDisplay, PtzHt)
```

C#

```
System.bool GetPTZHeight2(  
    System.int FrameNumber,  
    System.int TolNumber,  
    out System.bool PtzDisplay,  
    out System.string PtzHt  
)
```

C++/CLI

```
System.bool GetPTZHeight2(  
& System.int FrameNumber,  
& System.int TolNumber,  
& [Out] System.bool PtzDisplay,  
& [Out] System.String^ PtzHt  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber
Frame number
TolNumber
Tolerance number
PtzDisplay

True if the projected zone tolerance is displayed in the GTol, false if not

PtzHt
Height of the projected tolerance zone

Return Value

True if the method executed successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::GetPTZHeight2](#).

Example

[Insert Gtol \(C#\)](#)
[Insert Gtol \(VB.NET\)](#)
[Insert Gtol \(VBA\)](#)

Remarks

This method:

- is valid only if this Gtol was created in a version of SOLIDWORKS earlier than 2022.
- returns the height of the projected tolerance zone as a string because it is a user-specified parameter that can be textual, numeric, or both types of data.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::SetPTZHeight2 Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0